

Second Partial

Fco. Vazquez

Tec de Monterrey

Miau

This is a block

Here is the text of the block

Class Activity: application of Critical Points

The height of a rocket is given by

$$h(t) = \frac{400t}{\ln(t+1)}$$

The energy dissipated in a resistor is given the function

$$P(R) = \frac{100R}{(R+r)^2}$$

where R is the resistance and r is the internal resistance of the battery.

The velocity of a reaction is given by

$$v(s) = \frac{V_{\max} \cdot s}{K_m + s}$$

s is concentration.

The strenght of a pilar is given by

$$R(x) = k \cdot x \cdot \sqrt{d^2 - x^2}$$

Find the maximum point of resistance

Show that the critical point of $h(t)$ is $t = \frac{1}{e} - 1$ in $\left(v_o, v_o^2 + \frac{1}{e} \right)$